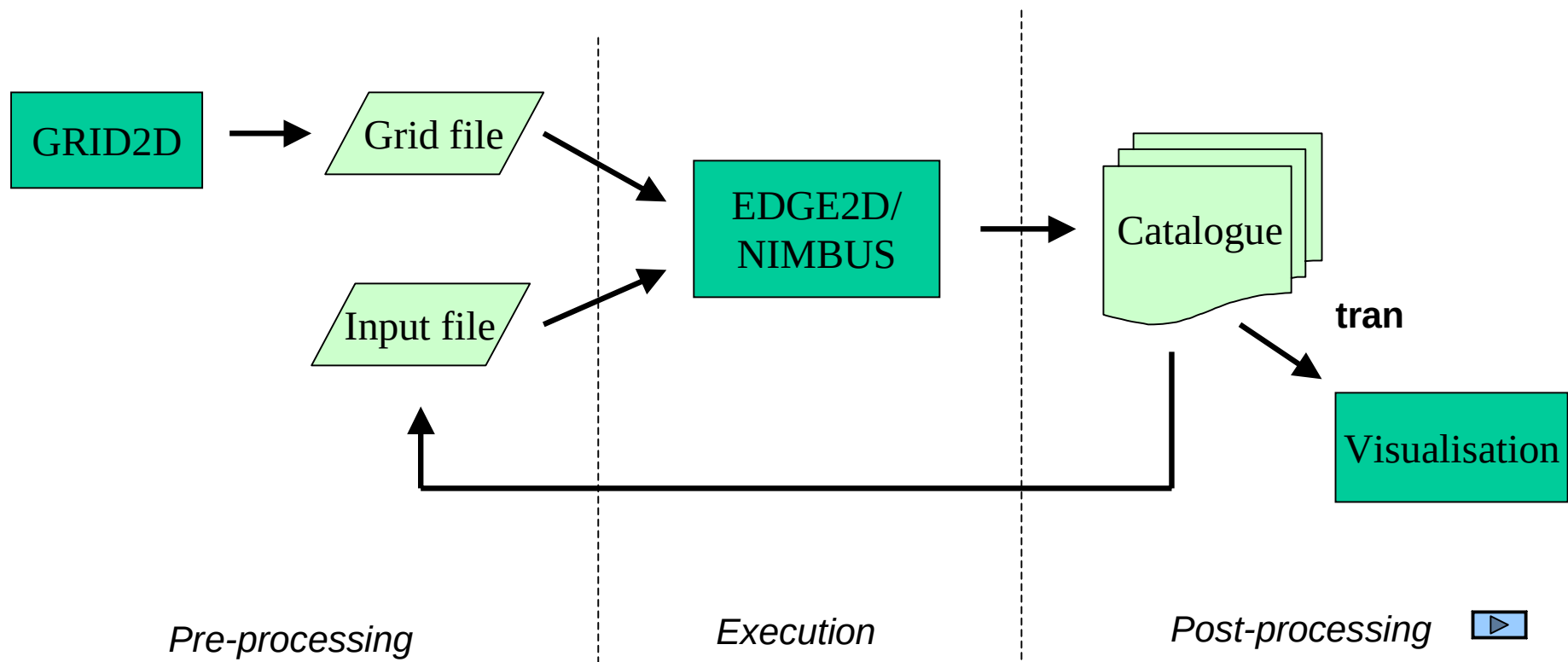


EDGE2D tools related to SOLPS

- **Developed / maintained by Code Management Group**
 - **Group Leader** : Vassili Parail (Ext. 4655)
 - **Developers** : Gerard Corrigan (Ext. 5189)
James Spence (Ext. 4865)
- **E-mail / subscribe - cmg-edge@jet.uk**

Overview

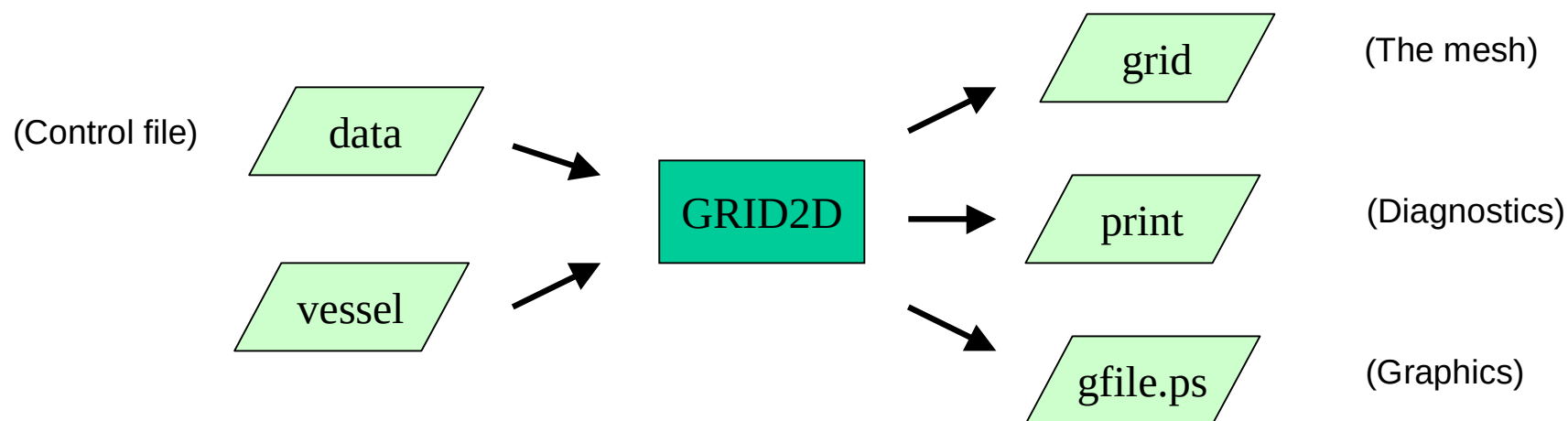


SOLPS relevance

- **Pre-processing**
 - GRID2D can create meshes in CARRE format
- **Post-processing**
 - Generic cataloguing / archiving system
 - B2.5 can create tran file $\Rightarrow$ visualisation tools

$\Rightarrow$ Common platform for comparisons with EDGE2D

GRID2D Overview



Grid generation with GRID2D

- **Installation**

- `/u/sim/sh/newsimuser` ($\Rightarrow$ modifies `.profile` and `.netrc`)

- **Documentation**

- Available on request
 - Format of control file
 - Format of vessel file
 - Central repository
 - Graphical output
 - Quick start guide

Grid generation with GRID2D ...continued

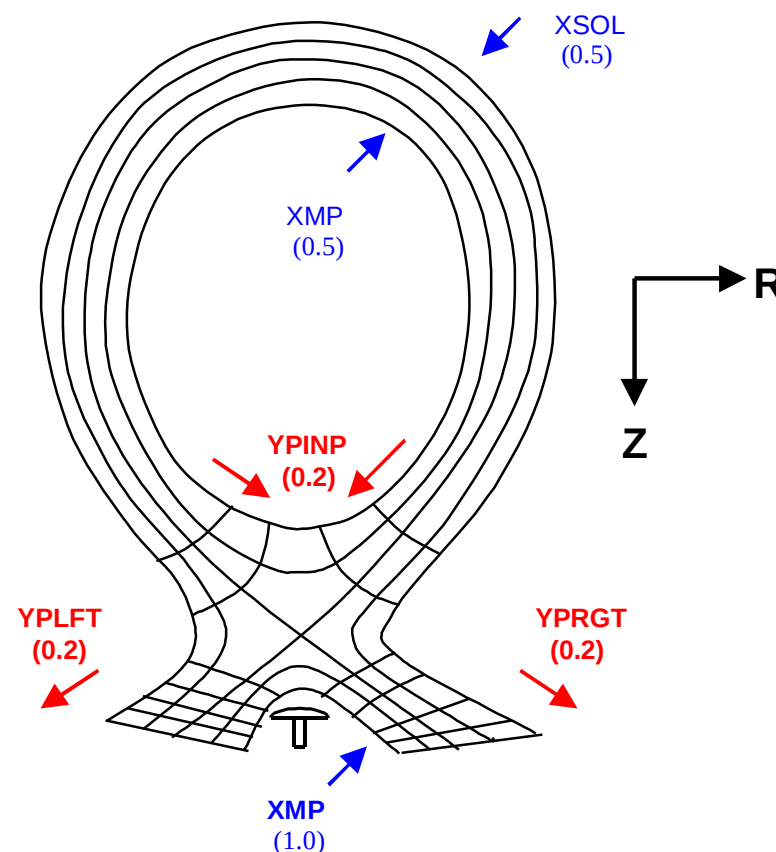
- **Prepare the input files**
 - Copy or create Vessel file
 - Edit control file **data**
- **Execute**
 - In directory with **data** use command **grid2d**
- **Check output**
 - View the graphical output with command **gv gfile.ps**
 - Check the diagnostics file **print** for error reports



Grid2D control file details (1)

- **&NPUT - key namelist variables**

LB2GRID		- create CARRE grid file
NBUFLE	}	- define the septum
RBUFLE		
ZBUFLE		
NBUFX		
RBUFX		
ZBUFX		
XMP,	XMP	- Core ring specification
XSOL,	XSOL	- SOL ring specification
XPRIG,	XPRIG	- Private region ring specification
YRGT,	YRGT	- Outer divertor row specification
YLFT,	YLFT	- Inner divertor row specification
YINP	YINP	- Row specification outside divertor



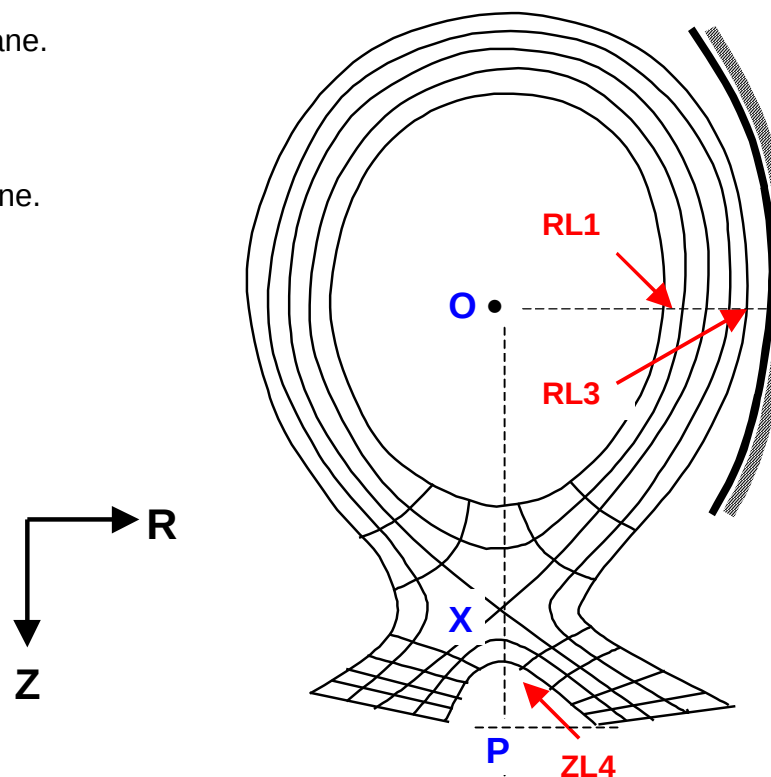
Grid2D control file details (2)

- **&NPUT - key namelist variables cont'd**

L1 - R of innermost core ring on mag. mid-plane.
(0.0 $\Rightarrow$ 1 cm core)

L3 - R of outermost core ring on mag. mid-plane.
(0.0 $\Rightarrow$ extend to vessel wall)

L4 - Z of outermost private ring along **OX**
(0.0 $\Rightarrow$ 1/3 **XP**)



Grid2D control file details (3)

- **&SHOT - key namelist variables**

ISHOT - Shot number
TSHOT - Time slice

IGO - Flush data source flag
e.g. +15 = EFIT PPF
-2 = File (in Flush specified format)

UID - PPF userid
ISEQ - PPF sequence number
DDA - DDA name for PPF data
CBANK - File name (for IGO = -2 option)

Grid generation with GRID2D ...continued

- **Common problems**

- Vessel conflict



- Septum conflict



- Double nulls

- Error $\Rightarrow$ “. . . more than 2 intersections”
 - Cure $\Rightarrow$ Try reducing RL3 to make SOL boundary inside 2nd separatrix

Still fails $\Rightarrow$ Contact CMG (James Spence)

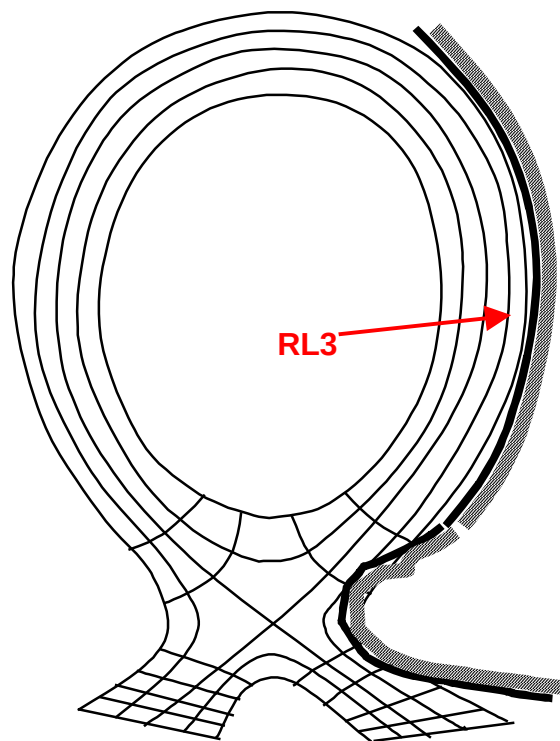
Overview

Grid2D common problems (1)

- **Vessel conflict**

Error $\Rightarrow$ “. . more than 2 intersections”

Cure $\Rightarrow$ Reduce RL3 and/or
adjust vessel

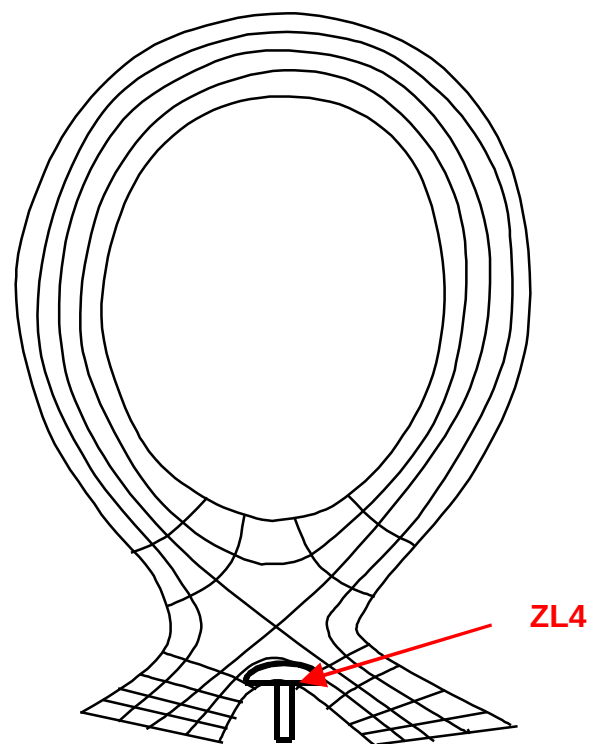


Grid2D common problems (2)

- **Septum conflict**

Error $\Rightarrow$ No Grid2d error.
Check graphical output for X-point position.

Cure $\Rightarrow$ Reduce ZL4 if possible.

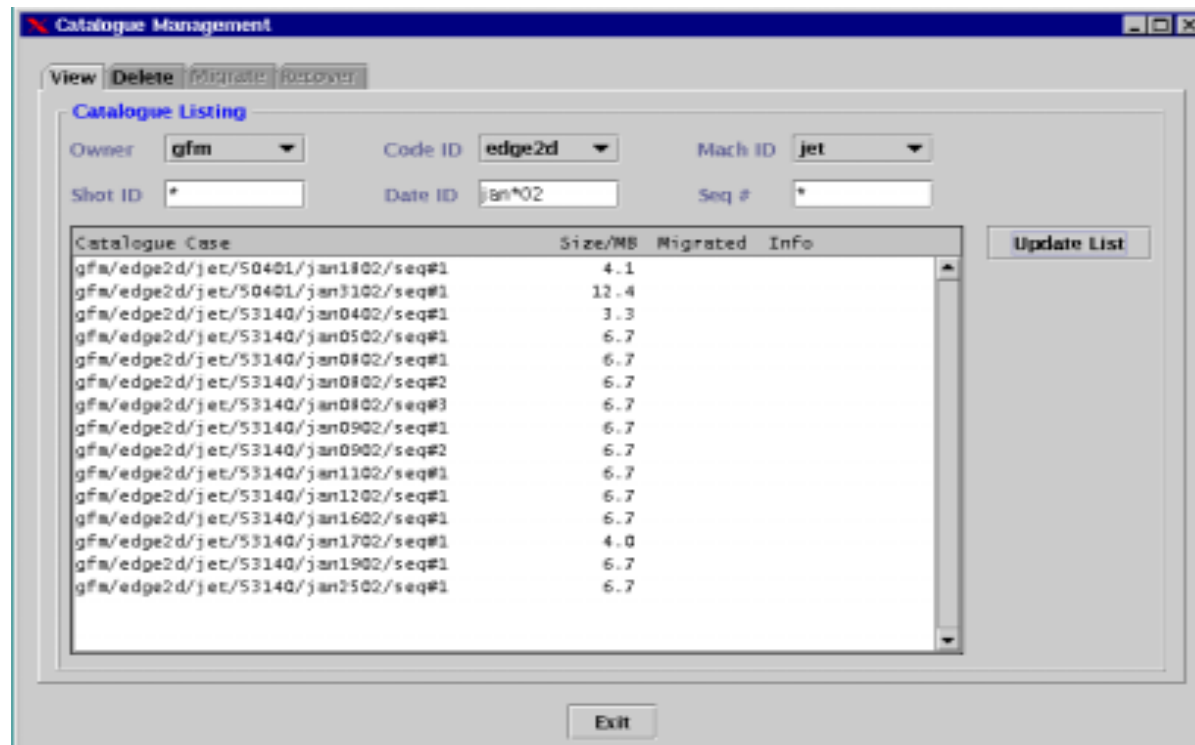


Post-processing - the catalogue system

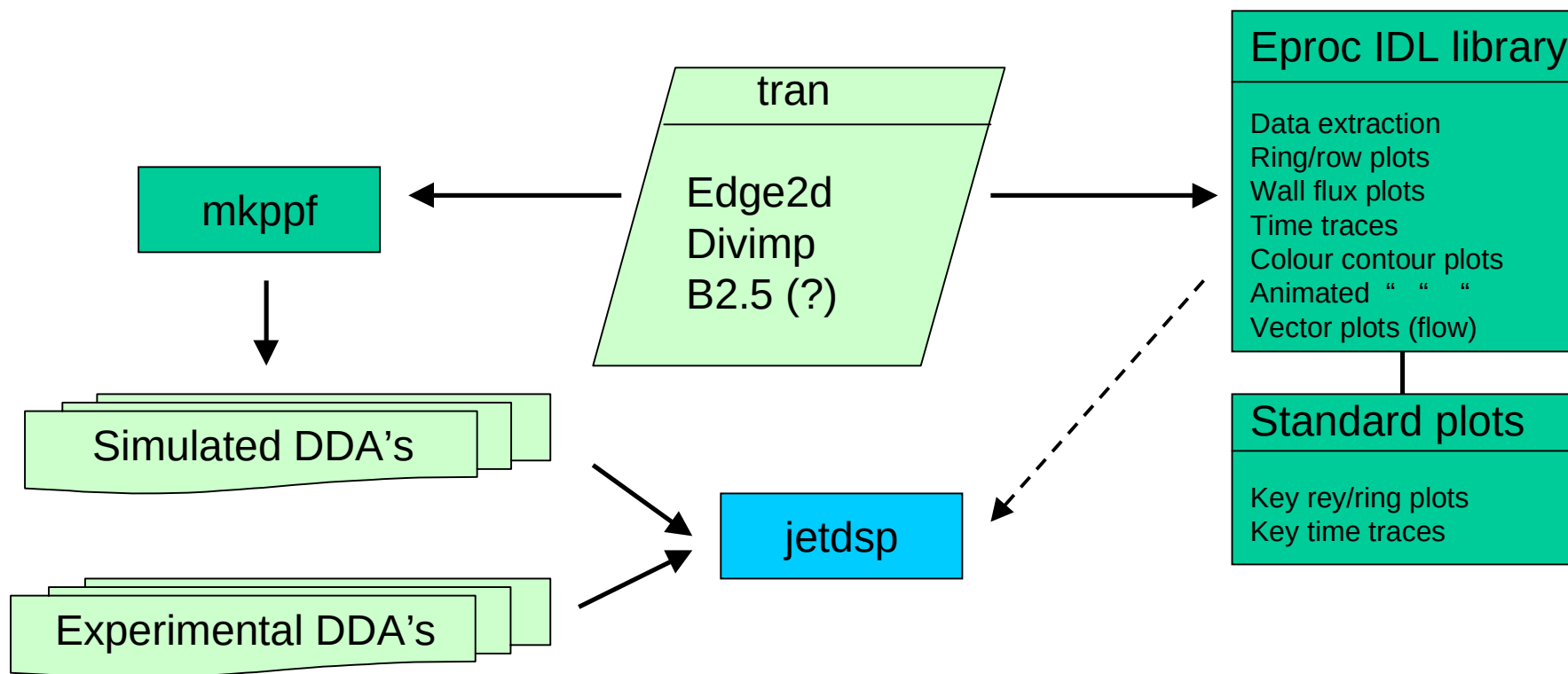
- **Generic system**
 - Configurable (e.g. Edge2d, Jetto, Coconut, Solps . . .)
- **The **catalog** command**
 - Name : <owner> <code> <machine> <shot> <date> <seq#>
 (e.g. gcor edge2d jet 53140 jan0102 seq#5)
 - Copies nominated input and output files
- **The **retrieve** (or **getrun**) commands**

Post-processing - the catalogue system

- The **catman** command



Post-processing - data visualisation tools



Summary

Tools originally developed for CMG codes can be used with SOLPS.

These provide :-

- **Common platform for comparing between different Edge codes and JET data**
- **A file management system for storage of simulations on JAC and sharing results with other users**